

Overland Erosion: From Rainfall Impact to Watershed Sediment Yield

Executive Summary

Overland erosion represents the critical interface between hydrology and sediment transport, completing a systemic cycle of rainfall, runoff, erosion, transport, and deposition. This document synthesizes the physical mechanics of soil detachment and movement, the empirical and process-based models used to predict soil loss, and the scaling factors required to estimate sediment yield at a watershed level.

The central framework for modern erosion analysis is the balance between rainfall-driven detachment (D_c) and flow-driven transport capacity (T_c), where total erosion is governed by the smaller of these two processes. While the Universal Soil Loss Equation (USLE) remains the foundational empirical model for agricultural and plot-scale analysis, modern management increasingly relies on process-based simulations such as the Water Erosion Prediction Project (WEPP) to account for deposition, event dynamics, and complex routing.

Understanding these processes is critical for engineering design, particularly as climate change amplifies both rainfall intensity and runoff shear stress, leading to increased sediment loads in reservoirs and stormwater infrastructure.

1. Physical Processes of Overland Erosion

Erosion is a multi-stage process occurring within a broader hydrological system:

Rainfall → Infiltration → Runoff → Erosion → Transport → Deposition

1.1 Types of Overland Erosion

Erosion manifests in four distinct stages, each increasing in severity and channel definition.

Splash Erosion

- Caused by raindrop impact detaching soil particles.
- Occurs when rainfall kinetic energy exceeds soil cohesion and adhesion.
- Highly dependent on rainfall intensity and soil type.

Sheet Erosion

- Characterized by thin-layer overland flow.
 - Selectively removes fine soil grains and transports them downslope.
 - Over time this process can form an **armor layer**, where coarser particles accumulate at the surface.
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Rill Erosion

- Development of small, narrow, well-defined channels caused by concentrated runoff.
 - Early-stage erosion process.
 - Can usually be corrected through normal tillage.
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Gully Erosion

- Severe degradation where rills enlarge into deep, steep-sided channels, typically deeper than approximately 300 mm.
- Unlike rills, gullies cannot be removed by normal tillage.
- Can destroy agricultural land and infrastructure.

Prevention generally requires maintaining vegetation cover greater than approximately 70%.

1.2 Conceptual Framework: Detachment versus Transport

Modern erosion modeling uses an analogy to sediment transport capacity and defines erosion as the minimum of two limiting factors:

1. Detachment (D_c): rainfall-driven particle detachment
2. Transport capacity (T_c): flow-driven sediment transport capacity governed by:

$$\tau \approx \rho g h S$$

where:

- τ = shear stress
- ρ = water density

- g = gravitational acceleration
- h = flow depth
- S = slope

The process with the smaller capacity controls the overall erosion rate:

$$E = \min (D_c, T_c)$$

2. Quantitative Modeling of Erosion

2.1 Rainfall and Detachment Mechanics

Soil detachment is influenced by raindrop terminal velocity and rainfall kinetic energy.

Raindrop terminal velocity

$$V \approx 120d^{0.35}$$

where:

- V = terminal velocity
 - d = raindrop diameter
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Rainfall kinetic energy

$$KE = 916 + 330 \log i$$

where:

- KE = kinetic energy
 - i = rainfall intensity
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Ellison (1947) Relation

The detached soil weight is estimated as:

$$G \propto V^{4.33} d^{1.07} i^{0.65}$$

where:

- G = detached soil weight
- V = raindrop velocity
- d = raindrop diameter
- i = rainfall intensity

2.2 Universal Soil Loss Equation (USLE)

USLE estimates annual soil loss per unit area:

$$E = R K L S C P$$

Factor	Description	Influencing Variables
R	Rainfall erosivity	Rainfall intensity and frequency
K	Soil erodibility	Soil properties
LS	Topographic factor	Slope length and steepness
C	Vegetation cover	Crop sequence and residue management
P	Conservation practice	Contouring, terracing, strip cropping

Limitations of USLE

- Purely empirical and plot-scale
- Does not model deposition
- Does not include routing
- Does not represent event dynamics

2.3 Modern and Process-Based Models

To overcome USLE limitations:

RUSLE / RUSLE2

- Revised versions of USLE used for conservation planning

- Applicable across broader land uses such as:
 - construction
 - mining
 - forests

WEPP

Water Erosion Prediction Project:

- Continuous simulation model
- Includes:
 - climate
 - plant growth
 - hydrology
 - residue decomposition
- Computes both detachment and deposition on hillslopes

SWAT

Used for:

- long-term watershed hydrology
- sediment routing

SWMM

Used for:

- urban runoff routing
- buildup and washoff modeling

3. Watershed Sediment Yield

Sediment yield is the portion of eroded material that reaches a downstream control point.

3.1 Key Definitions

Gross Erosion

Total soil eroded from watershed surfaces.

Sediment Yield

Sediment delivered to the watershed outlet.

Sediment Delivery Ratio (SDR)

$$SDR = \frac{\text{Sediment Yield}}{\text{Gross Erosion}}$$

3.2 SDR Scaling and Estimation

Smaller watersheds generally have higher SDR values because there are fewer opportunities for deposition.

Larger watersheds generally have lower SDR values because of:

- floodplain storage
 - channel deposition
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Methods for estimating sediment yield (Y)

1. Rating Curve Method

$$Y = \int Q_s f(Q) dQ$$

2. Reservoir Method

$$Y = \frac{A}{E}$$

$$Y = (\text{Accumulated sediment volume} / \text{time}) / \text{Trap efficiency}$$

3. USLE + SDR Method

$$Y = E_{gross} \cdot SDR$$

4. Urban and Construction Site Erosion

Construction sites experience elevated erosion rates due to soil disturbance.

Surface conditions influence the practice factor P :

- High erosion:

$$P \approx 1.30$$

Compacted smooth surfaces

- Low erosion:

$$P \approx 0.10 - 0.20$$

Sediment basins with flocculants

In urban environments, emphasis shifts toward:

Buildup

Pollutants accumulate over time.

Washoff

Rainfall removes accumulated material, often showing:

- rapid initial removal
 - subsequent decay behavior
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5. Case Study: Annual Sediment Yield Calculation

Example: Upland farm in Central Illinois (800 acres)

Step 1: Soil loss by USLE

Given:

$$\begin{aligned}R &= 200 \\K &= 0.33 \\LS &= 0.697 \\C &= 0.004 \\P &= 0.5\end{aligned}$$

Compute:

$$\begin{aligned}E &= 200(0.33)(0.697)(0.004)(0.5) \\&= 0.092 \text{ tons/acre/year}\end{aligned}$$

Step 2: Gross erosion

$$\begin{aligned}E_{gross} &= 0.092 \times 800 \\&= 73.6 \text{ tons/year}\end{aligned}$$

Step 3: Apply SDR

For:

$$\begin{aligned}&800 \text{ acres} \\SDR &\approx 0.417\end{aligned}$$

Step 4: Final sediment yield

$$\begin{aligned}Y &= 73.6(0.417) \\&= 30.7 \text{ tons/year}\end{aligned}$$

6. Engineering and Environmental Implications

6.1 Climate Change Amplification

Climate change acts as a multiplier:

$$\text{Climate Change} \Rightarrow R \uparrow, \tau \uparrow \Rightarrow E \uparrow$$

Higher rainfall intensity increases:

- R
- detachment

Higher runoff increases:

- τ
- transport capacity

Both sides of:

$$E = \min(D_c, T_c)$$

are amplified.

6.2 Design and Management

- **Vegetation control:** maintaining cover strongly influences C and P
- **Infrastructure impacts:** larger sediment loads reduce reservoir storage and stress stormwater systems
- **Regulatory needs:** design storms and erosion-control practices should incorporate changing rainfall intensity patterns